

# Calculus I Homework 4

Department of Mathematics

## Instructions

Show all work and simplify final answers. You may use a calculator for arithmetic only.

## Problems

1. Compute the limit, or state that it does not exist.

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = \boxed{4}$$

2. Find the derivative of the function.

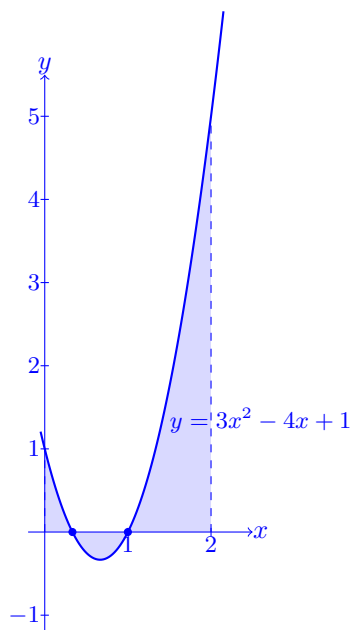
$$f(x) = x^3 \ln(x) - \frac{4}{x^2}$$

$$f'(x) = 3x^2 \ln(x) + x^2 + \frac{8}{x^3}$$

3. Compute the definite integral and sketch  $y = 3x^2 - 4x + 1$  with the area for the integral shaded.

$$\int_0^2 (3x^2 - 4x + 1) dx$$

$$\int_0^2 (3x^2 - 4x + 1) dx = \left[ x^3 - 2x^2 + x \right]_0^2 = (8 - 8 + 2) - 0 = \boxed{2}$$



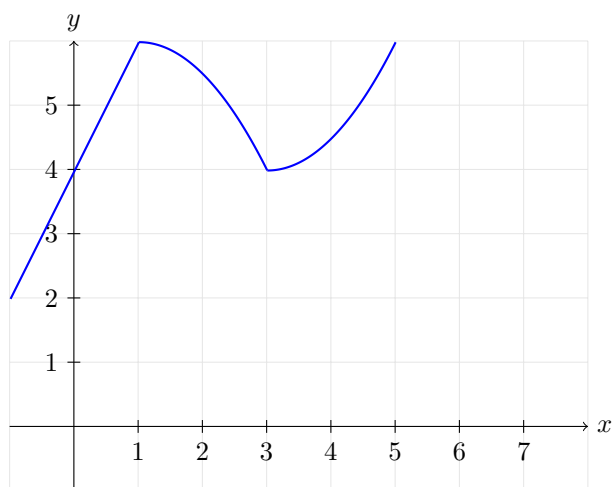
4. Let  $f(t)$  be defined by

$$f(t) = \begin{cases} 2 & 0 \leq t < 2, \\ 2 - t & 2 \leq t < 4, \\ t - 4 & 4 \leq t \leq 6. \end{cases}$$

Define

$$F(x) = \int_0^x f(t) dt.$$

On the axes below, sketch the graph of  $F(x)$  for  $0 \leq x \leq 6$ .



5. Use the Fundamental Theorem of Calculus to find the derivative.

$$G(x) = \int_1^x \sqrt{1+t^4} dt$$

$$\begin{aligned} G'(x) &= \sqrt{1+x^4} \cdot \frac{d}{dx}(x) \\ &= \boxed{\sqrt{1+x^4}} \end{aligned}$$